

Joshua Moses

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SUMMARY

Full Stack Developer skilled in building scalable, high-performance applications with a strong foundation in system design. Delivered production-grade projects spanning AI integration, cloud deployment, queue-based processing, and automation.

SKILLS

Languages	JavaScript, Java, Python	Databases	PostgreSQL, Redis, DynamoDB
Frontend	React + Vite, CSS, HTML	Cloud/DevOps	AWS, Docker, Vercel, GitHub, Kubernetes
Backend	Node.js, Express, FastAPI, Java	AI/ML	Ollama models, Amazon Bedrock, OpenCV, YOLOv8

EXPERIENCE

AI & Edge Robotics Engineer:

PEP Center, St. Joseph's Institute of Technology

2025 AUG-NOV

- Edge AI & Cloud: Collaborated and developed Nephele, a 5G-enabled edge AI robot using Raspberry Pi integrated with Amazon Web Services for real-time cloud processing and orchestration.
- Voice AI & LLMs: Implemented a voice interaction pipeline using OpenWakeWord, Whisper, and LLM-based responses via Ollama (Gemma) for efficient on-device inference.
- Automation & AI Features: Implemented intelligent modules including conversational voice assistant, 360° robotic navigation, interview assistant, attendance automation, and interactive teaching capabilities.

Backend developer Intern

Sysvine Technologies, Chennai

2024 JUNE - SEP

- Built RESTful APIs using Node.js and Express for real-time ingestion of multi-source healthcare registry data, serving 3 concurrent data pipelines with consistent uptime.
- Analyzed and integrated data validation and transformation middleware to ensure schema consistency across all ingested records, reducing malformed data incidents by ~60%.

PROJECTS

CodePro – Real-Time Coding Platform :

[Link](#)

- Code Execution: Built a scalable execution system integrating Judge0 with BullMQ and Redis for asynchronous job processing, handling up to 500 concurrent submissions with <2s evaluation latency.
- Distributed System : Implemented a queue-based architecture to enable concurrent processing, fault tolerance, and high system reliability under load.

Tech Stack : JavaScript, Node.js, React.js, BullMQ, Redis, Judge0, Docker

SentinelAI – AI Surveillance System :

[Link](#)

- Real-Time Video Processing & Alerting:: Designed a real-time, multi-tenant AI surveillance system using a modular microservice-style design that processes live RTSP CCTV streams from 10+ cameras simultaneously, performs zone-based object detection, and broadcasts alerts via WebSockets with intelligent escalation (WhatsApp + automated phone calls).
- Event-Driven Architecture: Established an event-driven detection pipeline with threat scoring, cooldown logic, per-camera alert thresholds, and MongoDB-backed incident persistence, implementing scalable system design principles such as worker isolation, async processing, RESTful APIs, real-time communication, and containerized deployment for production-grade monitoring.

Tech Stack - React, Fast API, OpenCV, Next.js, Docker, EC2, YOLOv8

ACHIEVEMENTS

Winner of Hack a Cloud 2024:

- MicroOps - A no code/ low code DevOps automation platform where developers new to DevOps can use a simple drag and drop interface for their web-application

Runner-up at AI ASCEND 2026 - Kyndryl :

- Uzhavan Connect - A voice-first, AI-powered agricultural marketplace that enables farmers to sell directly to consumers, access real-time mandi prices, and manage orders using simple voice interactions—even without internet.

EDUCATION

St. Joseph's College Of Engineering, Chennai

Bachelor's Of Technology in Information Technology(B.Tech IT)

2023-Present

CGPA: 8.4

Dominic Savio Matric Higher Secondary School, Chennai

HSC : 91%

2021-2023